Spectral Shape Optimization: A Mathematical Overview

How `douse` maximizes λ_k/λ_1 over polygons

douse project notes

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Abstract

This note is a self-contained mathematical tour of the shape-optimization problem `douse` solves: maximizing the ratio $\lambda_k(\Omega)/\lambda_1(\Omega)$ of Dirichlet–Laplacian eigenvalues over polygons of fixed perimeter. It states, without derivation, the Hadamard shape-derivative formulas for simple and repeated eigenvalues; works out what these formulas imply about the structure of optimal points of λ_k/λ_1, first when λ_k is simple and then when it sits inside a degenerate cluster; and closes by describing the optimization algorithms built on that structure and how they are wired together in `douse`. For a derivation-level treatment and implementation cross-references see `spectral_optimization.tex` in the same directory; this document is the map, that one is the territory.

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1 The problem

Let $\Omega \subset \mathbb{R}^2$ be bounded, open and connected, with Dirichlet spectrum

$$0 < \lambda_1(\Omega) < \lambda_2(\Omega) \le \lambda_3(\Omega) \le \cdots, \qquad -\Delta u_j = \lambda_j u_j \text{ in } \Omega, \quad u_j = 0 \text{ on } \partial\Omega,$$

eigenfunctions taken $L^2(\Omega)$-orthonormal. We study the functional

$$F(\Omega) = \frac{\lambda_k(\Omega)}{\lambda_1(\Omega)}, \qquad k \ge 2 \text{ fixed}, \tag{1}$$

over the class of N-gons of fixed perimeter, for a chosen N. Two structural facts set the stage.

Fact 1 (Krahn)**.** *λ_1 is always simple for connected Ω: its eigenfunction can be taken positive, and two orthogonal positive functions cannot coexist.*

Fact 2 (Payne–Pólya–Weinberger, proved by Ashbaugh–Benguria)**.** *$\lambda_2/\lambda_1 \le (j_{1,1}/j_{0,1})^2 = 2.5387340\ldots$ for every bounded connected planar Ω, with equality only for the disk.*

Fact 1 is the reason the *denominator* of F is always smooth: whatever happens to λ_k, λ_1 never participates in a degeneracy. Fact 2 bounds F itself when $k = 2$ and, more importantly here, gives a certificate: any computed $F > 2.5387$ is a bug, not a discovery.

Because $\Omega \mapsto c\Omega$ sends $\lambda_j \mapsto c^{-2}\lambda_j$, F is scale invariant. The perimeter constraint is therefore pure gauge-fixing — it removes a redundant degree of freedom from the parametrization without adding content to the problem. (This is unlike, say, $\min \lambda_1$, where the same constraint *is* the problem.) An N-gon's spectrum also ignores translation, rotation and reflection, so `douse` parametrizes the shape by a vector $p \in \mathbb{R}^{2N-4}$ of free vertex coordinates after fixing that gauge; the details are geometric bookkeeping and are not needed below beyond the one fact that $p \mapsto \Omega(p)$ is a smooth, locally invertible map on an open (but not all of $\mathbb{R}^{2N-4}$) admissible set.

2 The Hadamard formulas

Everything that follows rests on how λ_j responds to a perturbation of the domain. Let V be a velocity field on $\partial\Omega$ (only its normal component $V \cdot n$ matters) and let Ω_t be the domain swept out by transporting $\partial\Omega$ along V.

Fact 3 (Hadamard, simple eigenvalue)**.** *If λ is a simple eigenvalue with $\|u\|_{L^2(\Omega)} = 1$, then $t \mapsto \lambda(\Omega_t)$ is differentiable at $t = 0$ and*

$$\left.\frac{\mathrm{d}\lambda}{\mathrm{d}t}\right|_{t=0} = -\int_{\partial\Omega} \left(\frac{\partial u}{\partial n}\right)^2 (V \cdot n)\,\mathrm{d}s. \tag{2}$$

Fact 4 (Hadamard, repeated eigenvalue)**.** *If λ has multiplicity m with an orthonormal eigenbasis $u_1, \ldots, u_m$ for its eigenspace, define the symmetric $m \times m$ matrix*

$$M(V)_{ij} = -\int_{\partial\Omega} \frac{\partial u_i}{\partial n}\frac{\partial u_j}{\partial n}(V \cdot n)\,\mathrm{d}s. \tag{3}$$

Then the m eigenvalue branches bifurcating from λ under the perturbation $t \mapsto \Omega_t$ are, to first order in t, the eigenvalues of $M(V)$ — i.e. each branch has a one-sided derivative at $t = 0$ equal to one eigenvalue of $M(V)$, the branches ordered to match.

We take both facts as given; see Henrot–Pierre or Delfour–Zolésio for proofs. What matters for everything below is their *shape* rather than their derivation: (2) is a linear functional of $V \cdot n$, so a simple eigenvalue is differentiable with an honest gradient; (3) is a *matrix-valued* linear functional of $V \cdot n$, so a multiple eigenvalue is not differentiable at all in the ordinary sense — what survives is a well-defined first-order model for the whole cluster together, and that model is linear in V even though no individual branch is.

Fact 3 applied to λ_1 is unconditional, by Fact 1 — $\nabla_p \lambda_1$ exists everywhere on the (open) admissible set. Fact 4 applies to λ_k wherever the shape sits at (or near) a degeneracy, which for the regular N-gons that these problems tend to converge to is generic rather than exceptional: at a regular N-gon, every eigenvalue past λ_1 belongs to a cluster of size 1 or 2 (occasionally more, at special N), forced by the shape's dihedral symmetry D_N.

3 Optimal points of λ_k/λ_1

We now ask what (2)–(3) imply about a local maximum of F. Write $g_1 = \nabla_p \lambda_1$, which always exists.

3.1 λ_k simple

If λ_k is simple near p, F is differentiable there with

$$\nabla_p F = \frac{g_k\, \lambda_1 - \lambda_k\, g_1}{\lambda_1^2}, \qquad g_k = \nabla_p \lambda_k, \tag{4}$$

an ordinary quotient rule. A local maximum in the interior of the admissible set is characterized, to first order, by the single vector equation $\nabla_p F = 0$, equivalently

$$\lambda_1\, g_k = \lambda_k\, g_1, \tag{5}$$

i.e. the two shape gradients are parallel with a ratio fixed by the current value of F itself. Geometrically: at a critical point of F, the boundary velocity field that most increases λ_k is (up to sign and scale) the same field that most increases λ_1 — there is no direction that helps one without proportionally helping the other. This is a codimension-1-worth-of-conditions statement in each of the $2N - 4$ parameter directions and is otherwise unremarkable: it is exactly what one gets by differentiating any smooth ratio, with the geometric content entirely loaded into what g_1 and g_k *are*. Second-order conditions (a negative-semidefinite Hessian of F restricted to the admissible tangent space) then distinguish maxima from saddle points in the usual way, and are not special to this problem.

The interesting mathematics starts precisely where (5) stops applying: at the shapes these optimizations actually converge to, λ_k is essentially never simple.

3.2 λ_k degenerate: the directional derivative

Suppose λ_k sits in a cluster of multiplicity m, occupying position $j \in \{0, \dots, m-1\}$ when the cluster's members are listed in increasing order (so $j = 0$ is the bottom of the cluster, $j = m-1$ the top). By Fact 4, the one-sided directional derivative of λ_k in a parameter direction d is $\sigma_j(M(d))$, the j-th ordered eigenvalue of the affine matrix function $M(d) = \sum_i d_i M^{(i)}$ built

from (3) applied to the $2N-4$ coordinate velocity fields. Combining with the (always-defined) derivative of λ_1, the one-sided directional derivative of F itself is

$$\mathrm{d}F(d) = \frac{\sigma_j(M(d))\,\lambda_1 - \lambda_k\,(g_1 \cdot d)}{\lambda_1^2}. \tag{6}$$

(When the cluster's members are not exactly equal but merely close, the correct model replaces $M(d)$ by $\Lambda + M(d)$ with $\Lambda = \mathrm{diag}(\lambda_\ell - \lambda_k, \dots)$ recording the actual splitting; this refines (6) without changing anything said below, and is taken up in §4.)

Equation (6) is positively homogeneous of degree 1 in d (since σ_j is), so F is directionally, but not (in general) Gateaux, differentiable at p: $\mathrm{d}F(-d) \neq -\mathrm{d}F(d)$ whenever σ_j is genuinely nonlinear in d, which happens exactly when $0 < m \leq n_p$ and j is not both the top and bottom of a singleton cluster. F has a *kink* at every such shape, the multivariate analogue of $|x|$ at $x = 0$: it is the norm of the shape gradient near the corner $V = 0$ of the eigenvalue branches that produces the kink, not any failure of the underlying spectral data to vary smoothly.

3.3 First-order optimality at a degeneracy

At a local maximum, no direction can increase F to first order:

$$\mathrm{d}F(d) \leq 0 \qquad \text{for every admissible direction } d. \tag{7}$$

Because σ_j is positively homogeneous, (7) is equivalent to its restriction to $\|d\| = 1$, and it is exactly this restricted problem — maximize $\mathrm{d}F$ over a sphere (or, with active linear constraints, a sphere cap) — whose *value* is what a numerical certificate has to test: p is first-order optimal iff that maximum is ≤ 0. The structure of that maximization, and hence of the certificate and of the ascent step that precedes it, depends entirely on where λ_k sits inside its cluster:

position j	$\sigma_j(M(d))$ as a function of d	consequence for (7)
$j = 0$ (bottom of cluster)	concave	concave maximization: a genuine, tractable optimum
$j = m-1$ (top of cluster)	convex	maximizing a convex function over a ball: the max sits on the boundary and the feasible set of *improving* directions is generically large unless a further reduction applies (§4)
$0 < j < m-1$ (interior)	neither	no closed form; genuinely nonconvex, though cheap to evaluate
$m = 1$ (simple)	linear	recovers §3.1

This dichotomy is the load-bearing fact of the whole theory: $\lambda_{\min}$ of a symmetric affine matrix function is concave and $\lambda_{\max}$ is convex, a classical and easy fact (each is an infimum, respectively supremum, of linear functions $u^\top M(d) u$ over unit vectors u), and it is what makes the bottom of a cluster qualitatively different from the top.

Which case actually occurs. At the regular N-gons that these optimizations converge to, the pattern (forced by D_N symmetry and confirmed numerically) is λ_1 simple, then clusters of size 1 or 2 alternating, with λ_2 always the *bottom* of its pair. Maximizing λ_2/λ_1 therefore lands in the tractable concave row for every N; the higher ratios $\lambda_4/\lambda_1, \ldots$ can land in any row depending on N and k, which is why `douse` needs all three branches of §4 rather than only the concave one.

3.4 A worked degeneracy: when the first-order test is vacuous

It is worth seeing explicitly that (7) can hold identically, carrying no information, which is a genuine phenomenon and not a numerical artifact. Take $k = 2$, $m = 2$ at a regular N-gon. Two facts hold there exactly:

1. $g_1 = 0$ — the regular N-gon minimizes λ_1 at fixed perimeter among N-gons (Polya–Szegő, $N = 3, 4$; conjectured/observed more generally), so the denominator's gradient vanishes at exactly the shapes where the numerator degenerates;

2. $\operatorname{tr} M(d) = 0$ for every d — the symmetric function $\lambda_2+\lambda_3$ is stationary there, a consequence of (3) and the shape's symmetry.

With $g_1 = 0$, (6) collapses to $\mathrm{d}F(d) = \sigma_0(M(d))/\lambda_1 = \lambda_{\min}(M(d))/\lambda_1$. Because $M(-d) = -M(d)$, the sign condition $\lambda_{\min}(M(d)) \le 0$ for all d is equivalent to $M(d)$ never being positive definite, and for a 2×2 block exactly to $\det M(d) \le 0$ for all d. The traceless condition forces $M(d) = \left(\begin{smallmatrix} \alpha & \beta \\ \beta & -\alpha \end{smallmatrix}\right)$ with α, β linear in d, so $\det M(d) = -\alpha^2 - \beta^2 \le 0$ *identically* — (7) is satisfied automatically, for every regular N-gon, and certifies nothing about optimality: the quadratic form behind it is negative semidefinite of rank at most 2 no matter how many free parameters n_p the shape has. The verdict is then decided at second order, on the null space where the first-order test is silent; only for $N = 3$ does that null space vanish (there $n_p = 2$ exactly matches the rank), making the equilateral triangle a genuine first-order *strict* local maximizer of λ_2/λ_1 among triangles.

The moral for the rest of the document: a first-order stopping test at a cluster is a real mathematical statement, not a formality, and it can degenerate into a tautology at exactly the symmetric points the optimization is trying to find. `douse` treats this as a fact to be checked (`certify.py`), not something the ascent algorithm itself can be trusted to notice.

3.5 Kinks that are geometric, not spectral

One further nonsmooth phenomenon deserves separating out because it looks like the eigenvalue degeneracy of §3.2 but has nothing to do with it: F also has kinks at shapes where the polygon itself degenerates, e.g. where a vertex becomes flat (an N-gon momentarily coinciding with an $(N-1)$-gon). There λ_k is simple and smooth in the shape, but the exponent $\nu = \pi/\omega$ governing the corner behavior of the eigenfunction crosses 1 as the interior angle ω passes through π, and the two sides (reentrant, $\nu < 1$; convex, $\nu > 1$) are different analytic problems meeting at a point. The result is again a concave kink with finite, unequal, nonzero one-sided derivatives, and again a first-order model cannot land on it. This is a fact about the *parametrization* intersecting a fact about corner singularities, not about the multiplicity structure of §3.2, and `douse` handles the two with genuinely different machinery (§4).

4 Optimization algorithms

Sections 2–3 describe the local model available at a point p: a gradient when λ_k is simple, a matrix pencil when it is not, always with g_1 available. `douse` turns that model into an ascent method by a standard device — trust-region maximization of the local model, followed by an accept/reject test against the true objective — specialized in three ways to the structure above.

4.1 Trust-region ascent, in outline

At iterate p with radius Δ, propose $d^\star = \arg\max_{\|d\|\le\Delta}(\text{local model of } \mathrm{d}F)$, evaluate $F(p+d^\star)$ exactly (one eigensolve), and accept or shrink according to the usual ratio test $\rho = (F(p+d^\star) - F(p))/(\text{predicted gain})$. Because the local model is positively homogeneous of degree 1 (§3.2), its optimal value over $\|d\| \le \Delta$ scales linearly in Δ; the stopping test is therefore not the raw optimal value but *criticality* $= (\text{optimal gain})/\Delta$, the best achievable directional derivative, which is the quantity (7) actually tests and which is scale-free.

4.2 The smooth phase: BFGS/SR1 inside an exact trust-region solve

Away from any degeneracy (§3.1), the local model is linear and its maximizer over a ball is trivial ($d \propto g_k$ or ∇F), which wastes the curvature information a run accumulates for free along the way. `douse` instead fits quadratic models $g\cdot d + \frac{1}{2}d^\top B d$ to $-\lambda_1$ and to the (possibly still-active) cluster mean, using symmetric-rank-one updates (SR1, chosen over BFGS because neither scalar is known to be convex along the ascent path) built from consecutive *accepted* smooth gradients, and solves the resulting trust-region subproblem exactly via the More–Sorensen secular-equation characterization $d(\sigma) = -(B+\sigma I)^{-1}g$ — cheap here since the parameter dimension is at most a few dozen. The curvature chain is deliberately broken (not discarded) whenever the model switches into cluster mode, since a secant pair straddling a nonsmooth excursion compares two things that are not the same function.

4.3 The cluster phase: matching the subproblem to j

When λ_k is (nearly) degenerate, §3.3's three cases become three different subproblem solvers, all maximizing (6) over $\|d\| \le \Delta$:

- **Bottom of cluster ($j = 0$), concave.** For a 2×2 block the matrix inequality defining $\lambda_{\min} \ge t$ reduces to two scalar conditions on trace and determinant, solved as a small smooth program; equivalently (and more robustly, avoiding a degenerate constraint gradient at the natural starting point) $\lambda_{\min}$ of a 2×2 symmetric matrix is written directly as *linear minus a norm*, turning the subproblem into an unconstrained-looking concave program in d alone. For $m \ge 3$ blocks the honest semidefinite program is solved numerically.

- **Top of cluster ($j = m-1$), convex.** Maximizing a convex function over a ball puts the optimum on the sphere, but for $m = 2$ the identity $\lambda_{\max} = \ell\cdot d + \|Ld\|$ (with ℓ, L linear in d) lets the maximization over $\|d\| = \Delta \subset \mathbb{R}^{n_p}$ be exchanged for a maximization over the unit circle in $\mathbb{R}^2$: a one-dimensional search, exact. For $m \ge 3$ the same reduction becomes a maximization over the sphere S^{m-1}, handled by multi-start.

- **Interior of cluster ($0 < j < m - 1$).** Neither reduction applies, and the subproblem is genuinely nonconvex; but since evaluating the model at any trial d is a cheap $m \times m$ eigenvalue problem, it is simply maximized directly by multi-start local optimization, seeded from the two extreme-case solutions above. This branch carries no first-order certificate of its own, only empirical agreement with brute-force search.

Whichever branch produces the step, the criticality test of §3.3 is always evaluated against the exact model (6), so a heuristic interior step can still be checked against the certificate it does not itself supply.

Near, but not exactly at, a degeneracy the cluster is only ε-*active*: its members are close, not equal, and the correct affine model is $\Lambda + M(d)$ rather than $M(d)$ alone, with Λ the diagonal matrix of the actual (small) splitting. Treating a near-degeneracy as an exact one makes the existing splitting look free and produces a systematically over-optimistic predicted gain; the width of the active set is tied to the trust radius so that it neither under-clusters (which manufactures a fictitious O(1) gradient at a genuinely nonsmooth point) nor over-clusters indefinitely as the run converges.

4.4 Second-order refinement of the ratio itself

The model (6) linearizes numerator, denominator, and quotient all at once, which is more loss than necessary: the quotient's own curvature is exact arithmetic once the two pieces are in hand, so `douse` additionally forms

$$\widehat{F}(d) = \frac{\lambda_k + \sigma_j(\Lambda + M(d)) + \frac{1}{2} d^\top H_\mu d}{\lambda_1 + g_1 \cdot d + \frac{1}{2} d^\top H_1 d} - \frac{\lambda_k}{\lambda_1},$$

carrying SR1 curvature H_1, H_μ on the two genuinely smooth scalars — λ_1 and the cluster mean μ — while leaving the nonsmooth eigenvalue term σ_j at its exact, first-order-only, form (a second-order model of a kink is fiction). The linear/matrix-pencil steps above are retained as seeds for maximizing $\widehat{F}$, so the refinement can only improve on them, never lose ground.

4.5 Crossing geometric kinks

The kinks of §3.5 are not reachable by any local step, however the trust region is sized, because they sit at a discontinuity in the parametrization itself (a flat vertex, or a collapsed edge, correspond to different regions of p-space that a smooth path cannot connect). `douse` instead treats the two canonical degenerate configurations — exactly flattening a nearly-flat vertex, and deleting a very short edge — as discrete *trial moves*: propose the candidate shape, evaluate F there exactly, and accept it under the same improvement test used for an ordinary step. This is offered only when an ordinary step has stalled at very small radius near such a configuration, and it is what lets a run cross, for instance, from a pentagon pinned against a minimum-edge-length constraint to the true quadrilateral optimum it is degenerating toward.

4.6 Assembling λ_k/λ_1 maximization

Putting the pieces together, one iteration of `douse`'s spectral-ratio maximizer (`douse.cluster_opt.maximize_ratio`) does the following:

1. Solve for the current spectrum and detect whether λ_k is simple or belongs to an ε-active cluster, sizing the cluster width from the current trust radius.

2. Build the local model: g_1 always (Fact 3 applied to λ_1), and either g_k (simple case) or the pencil $\{M^{(i)}\}$ and splitting Λ (cluster case, Fact 4).

3. Dispatch to the matching trust-region subproblem of §4, refine with the second-order quotient model, and test the resulting step's criticality against the linear model; stop (and report an achieved *resolution*, not a bare pass/fail) if criticality is at or below tolerance, or if it has stopped changing as the radius shrinks — a signal that the floor has been reached is the accuracy of the eigenvalue solves themselves, not that the point is a first-order maximizer to arbitrary precision.

4. Otherwise take the step (subject to any geometric constraints, linearized into the same subproblem), evaluate the true ratio, and accept or shrink by the ratio test; if the step is instead rejected while the shape is pinned near a geometric kink, offer the discrete crossing moves of §3.5.

5. On termination, re-solve the eigenproblem once from a cold start at the reported optimum, since a continuation-tracked spectrum converging *onto* a degeneracy is exactly where tracking is least trustworthy, and report the invariants of §1–2 as a sanity check on the answer.

The upshot is that the algorithm's control flow mirrors the mathematics of §3 directly: every branch it can take corresponds to one row of the position-in-cluster table, and the two places it departs from a plain smooth trust-region method — the matrix-pencil subproblems and the discrete kink-crossing moves — are exactly the two places §3 identified λ_k/λ_1 as failing to be an ordinary smooth function of the shape.

Pointers

§2	`douse/derivative.py`; the underlying quadrature is `lappy`'s `hadamard_quad`
§3.2–§3.4	`douse/derivative.py` (`cluster_matrices`), `douse/certify.py`
§3.5	`douse/degeneracy.py`
§4	`douse/cluster_opt.py` (`local_model`, `trust_region_step`, `maximize_ratio`), `douse/quasi_newton.py`